

■ Data Visualization with Matplotlib & Seaborn

Yoohee Colleges

The Core Objective

Why do we visualize? Moving from raw data to actionable insights.

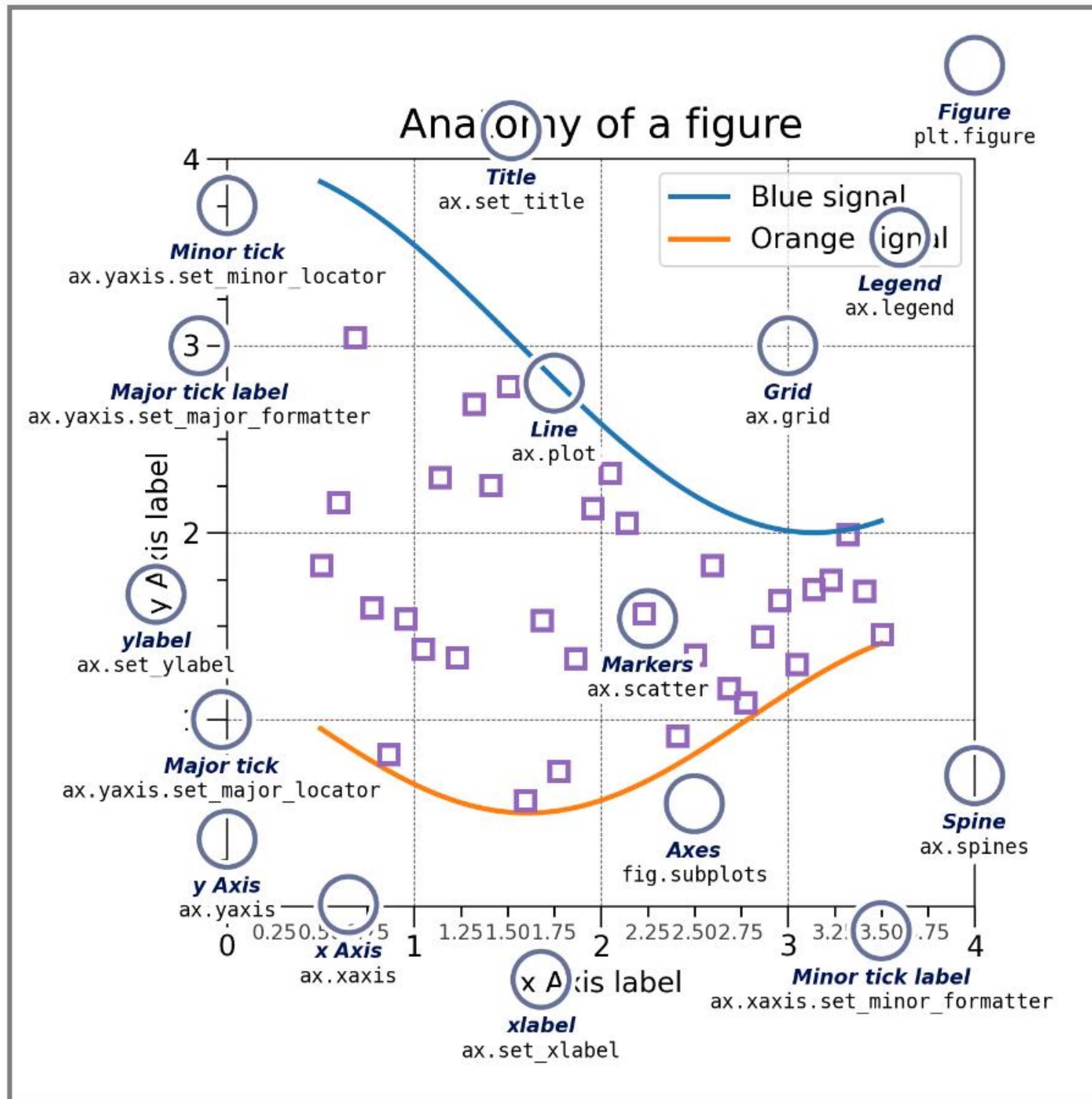
| The Python Visualization Stack

Python offers a powerful hierarchical stack for rendering graphics:

- **Matplotlib:** The low-level foundation providing total control over every pixel.
- **Seaborn:** A high-level interface built on Matplotlib, optimized for statistical patterns.
- **Integration:** Seamless compatibility with Pandas DataFrames and NumPy arrays.



Anatomy of a Matplotlib Plot



Key Components

- **Figure:** The top-level container (the whole page).
- **Axes:** The actual plot area where data is drawn.
- **Axis:** The number-line-like objects (X and Y).
- **Artists:** Everything you see on the figure (lines, text, markers).

| Standard Matplotlib Workflow



Create

Initialize a figure and axes using `plt.subplots()`. This creates the canvas for your data.



Plot

Use methods like `ax.plot()`, `ax.scatter()`, or `ax.bar()` to render data points.



Label

Customize with `set_title()`, `set_xlabel()`, and `legend()` for clarity.

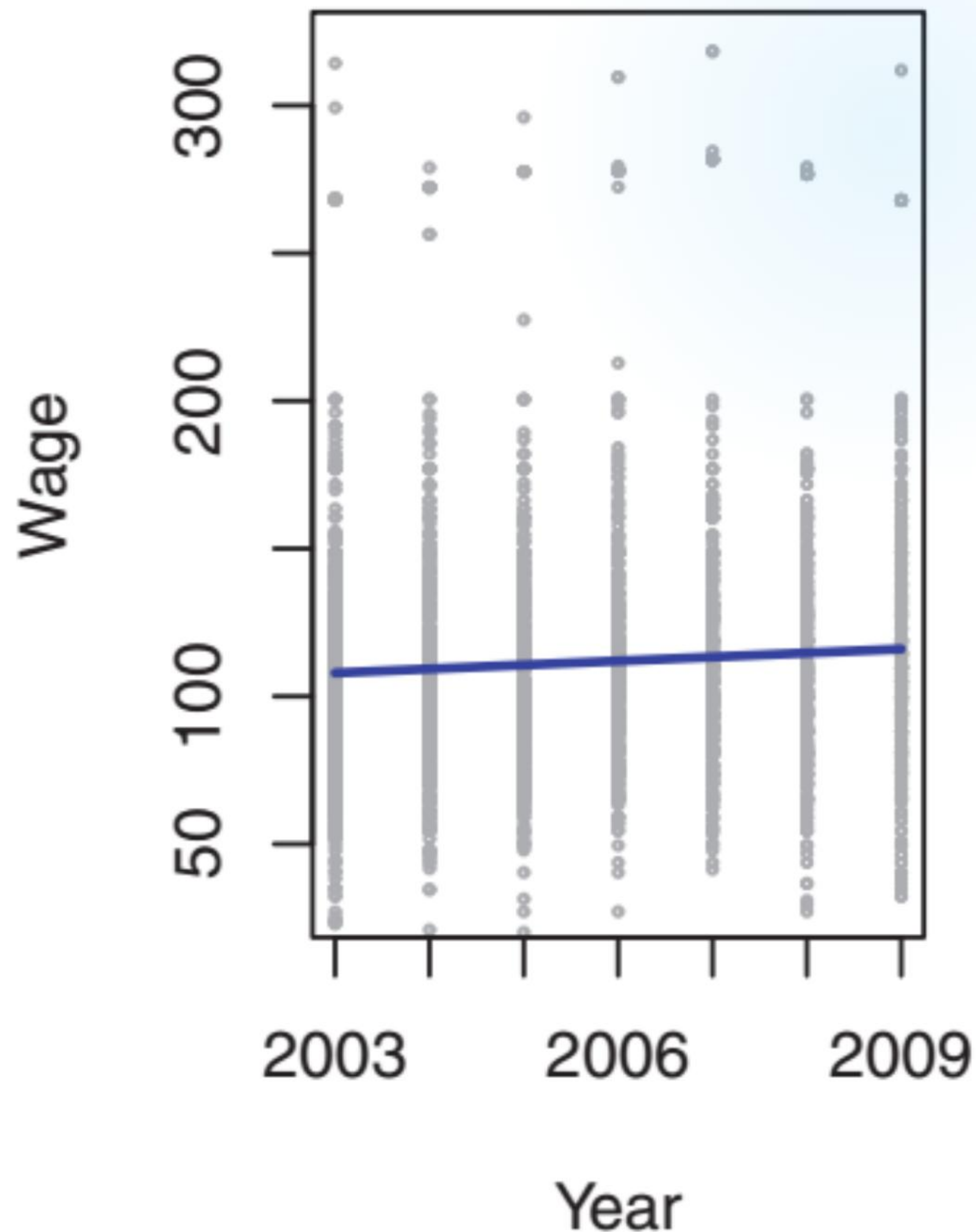
Enter Seaborn

Statistical Elegance

Seaborn takes the complexity of Matplotlib and wraps it in an opinionated, high-level API.

It is designed to "make a hard thing easy" by automating statistical estimation and providing beautiful default color palettes.

"If Matplotlib tries to make easy things easy and hard things possible, Seaborn tries to make a well-defined set of hard things easy too."



| Why Professionals Choose Seaborn



Pandas Native

Works directly with DataFrames. Just specify the column name for X, Y, and Color (Hue).



Auto Statistics

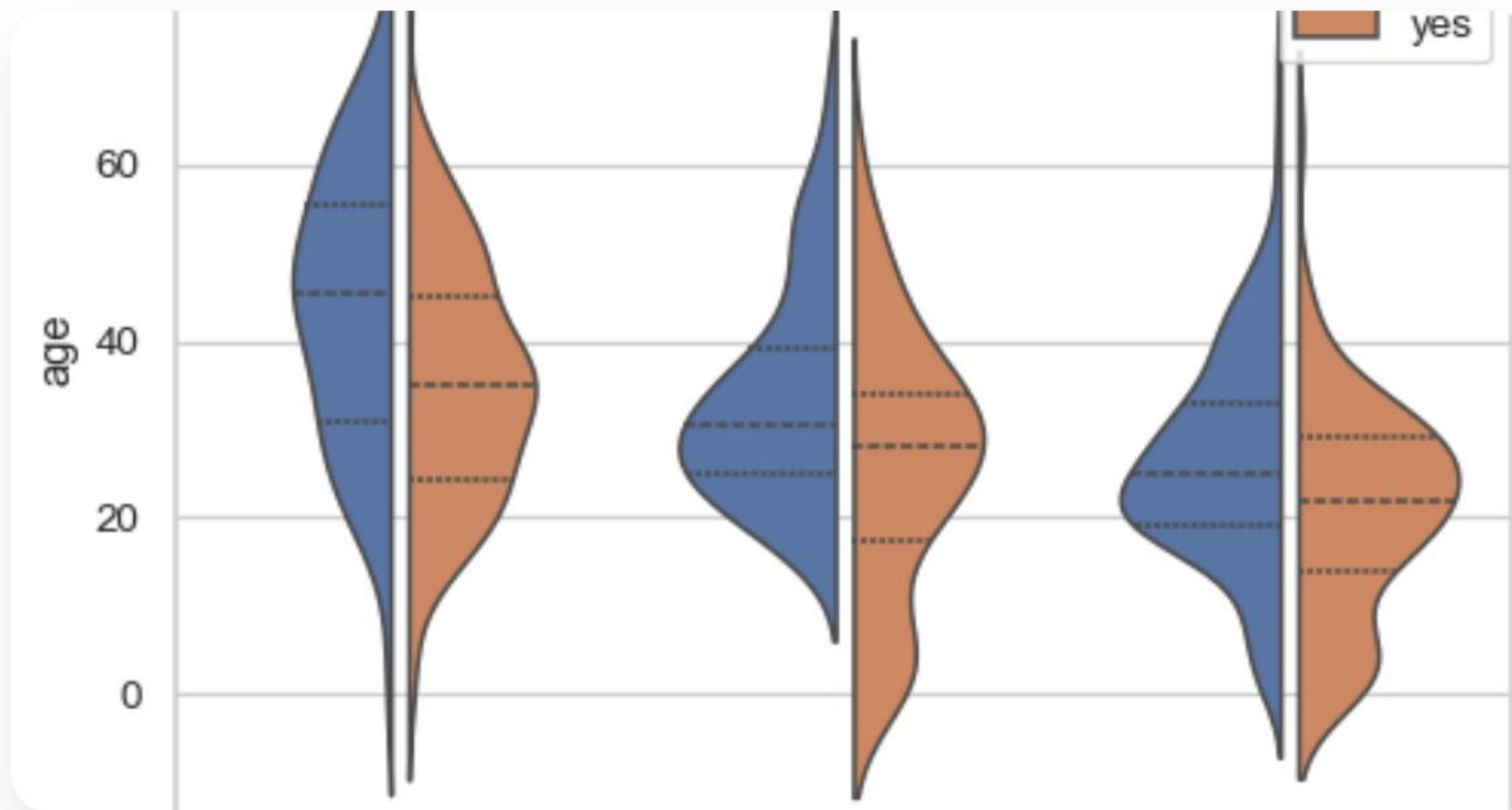
Automatically calculates confidence intervals, error bars, and regression fits.



Themes

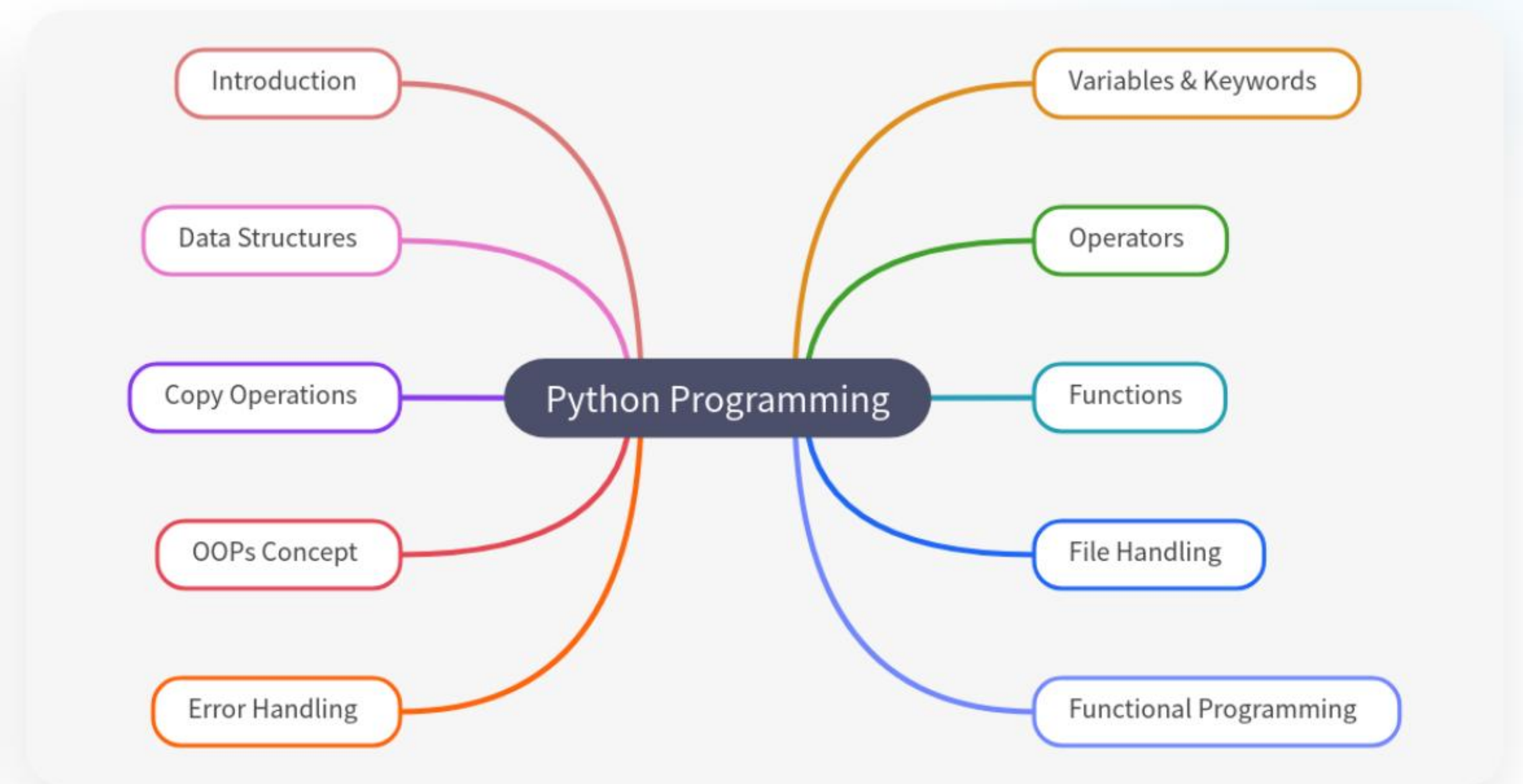
Built-in themes like darkgrid or ticks change the aesthetic with one line.

Understanding Data Spread



Distributions

Histograms & KDE plots reveal underlying probability density.



Aggregates

Boxplots and Violin plots identify outliers and quartiles.

Library Usage Trends



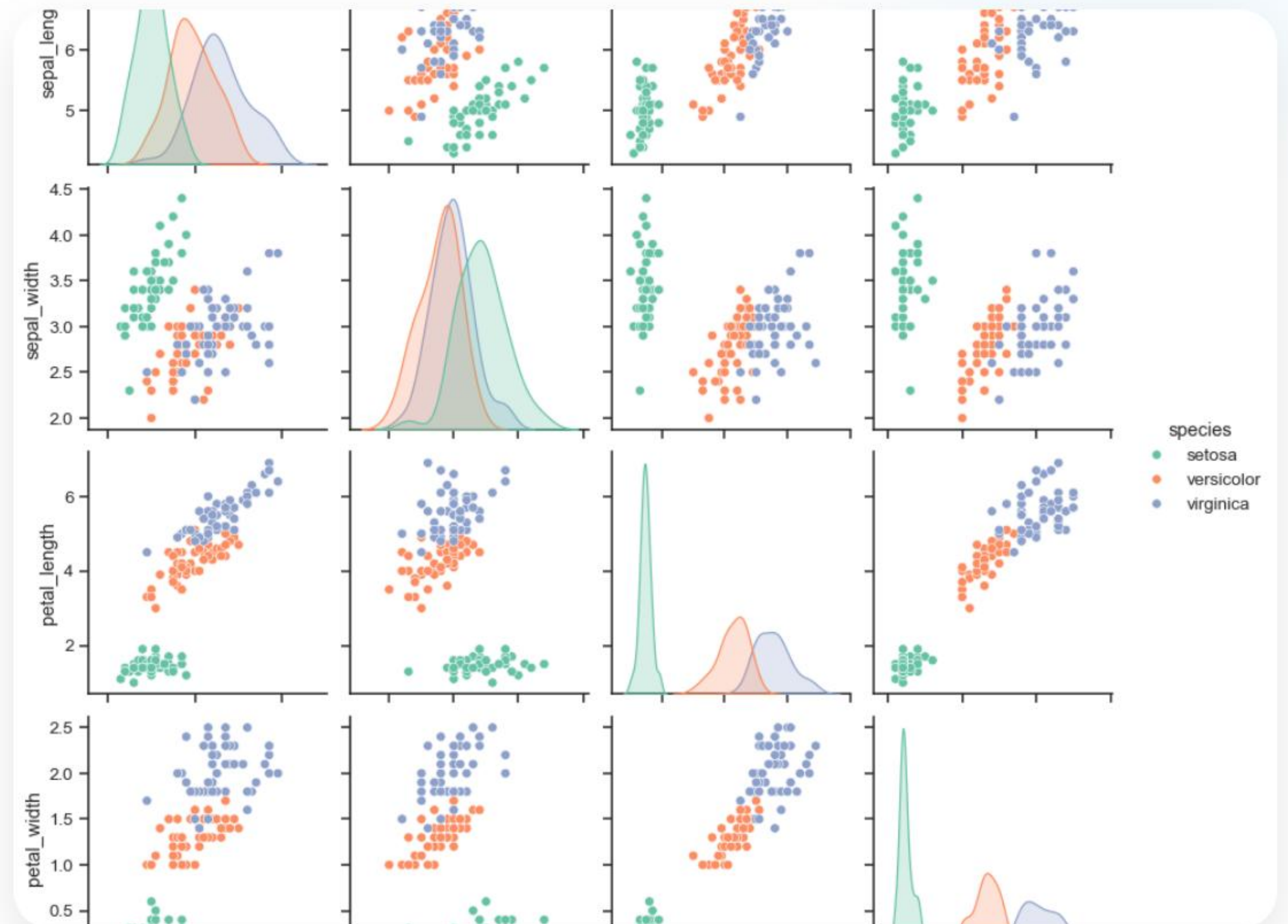
Survey data indicating user preference based on task complexity and visual requirement.

The Power of Multi-Plot Grids

PairPlots & FacetGrids

One of Seaborn's strongest features is the ability to visualize multidimensional relationships in a single command.

- **PairPlot:** Visualizes pairwise relationships across an entire dataset.
- **FacetGrid:** Maps a plot type across rows and columns based on category.



Feature Comparison

Feature	Matplotlib	Seaborn
Level of Abstraction	Low (Granular)	High (Functional)
Ease of Use	Moderate	High (Single functions)
Aesthetics	Basic Defaults	Premium Defaults
Statistical Tools	Manual	Automatic (Built-in)

Questions?

Thank you for your time. Start visualizing your data! 🙌🍷

 Documentation: matplotlib.org | seaborn.pydata.org